

Objectives

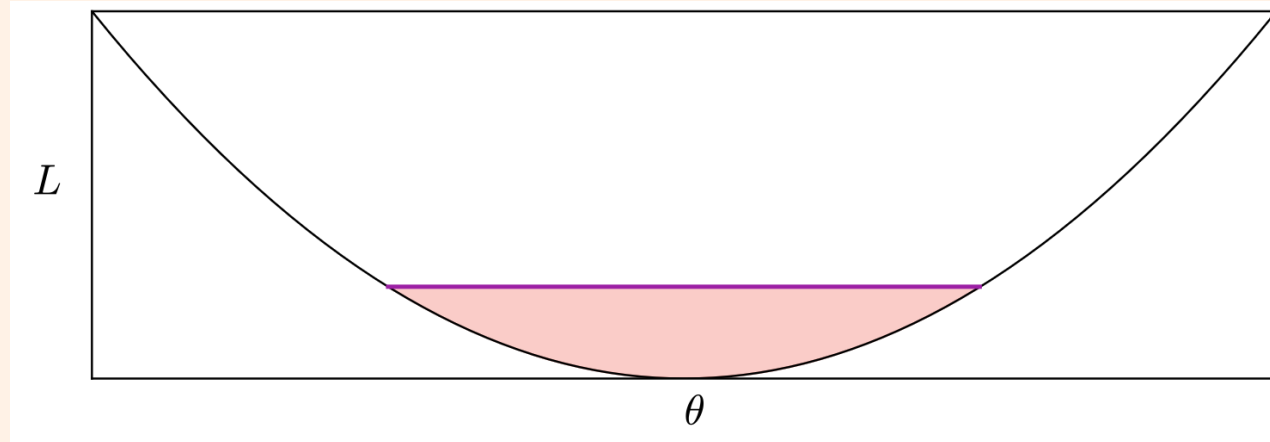
- Question the assumption that flatter minima always generalize better.
- Evaluate the reliability of common flatness measures: Volume ϵ -flatness, Hessian-based sharpness, and ϵ -sharpness.
- Demonstrate that these measures are not invariant to reparameterizations or input transformations, and that sharpness can be manipulated without affecting model predictions.
- Show that generalization claims based on flatness are unreliable unless the parameter space and input representation are clearly defined, due to underlying symmetries and non-identifiability in deep networks.

Notations and Definitions of flatness and sharpness

Definition 1. Given a tolerance $\epsilon > 0$, loss function L , and, $C(L, \theta, \epsilon)$ as the largest connected set containing θ such that:

$$\forall \theta' \in C(L, \theta, \epsilon), \quad L(\theta') < L(\theta) + \epsilon.$$

The volume of $C(L, \theta, \epsilon)$ is the ϵ -flatness defined at θ .



ϵ -flatness illustration around local minimum θ .

Definition 2. Let $B_2(\epsilon, \theta)$ be an Euclidean ball centered on a minimum θ with radius ϵ . Then, for a non-negative valued loss function L , the ϵ -sharpness will be defined as proportional to

$$\frac{\max_{\theta' \in B_2(\epsilon, \theta)} (L(\theta') - L(\theta))}{1 + L(\theta)}.$$

In the figure, the black curve shows the loss $L(\theta)$. The red region marks points where the loss is within ϵ of the minimum. If the height is ϵ , the width gives the **volume ϵ -flatness**. If the width is fixed at 2ϵ , the height reflects ϵ -**sharpness**. The **purple line** at the top represents $C(L, \theta, \epsilon)$, whose length corresponds to the flatness in 1D.

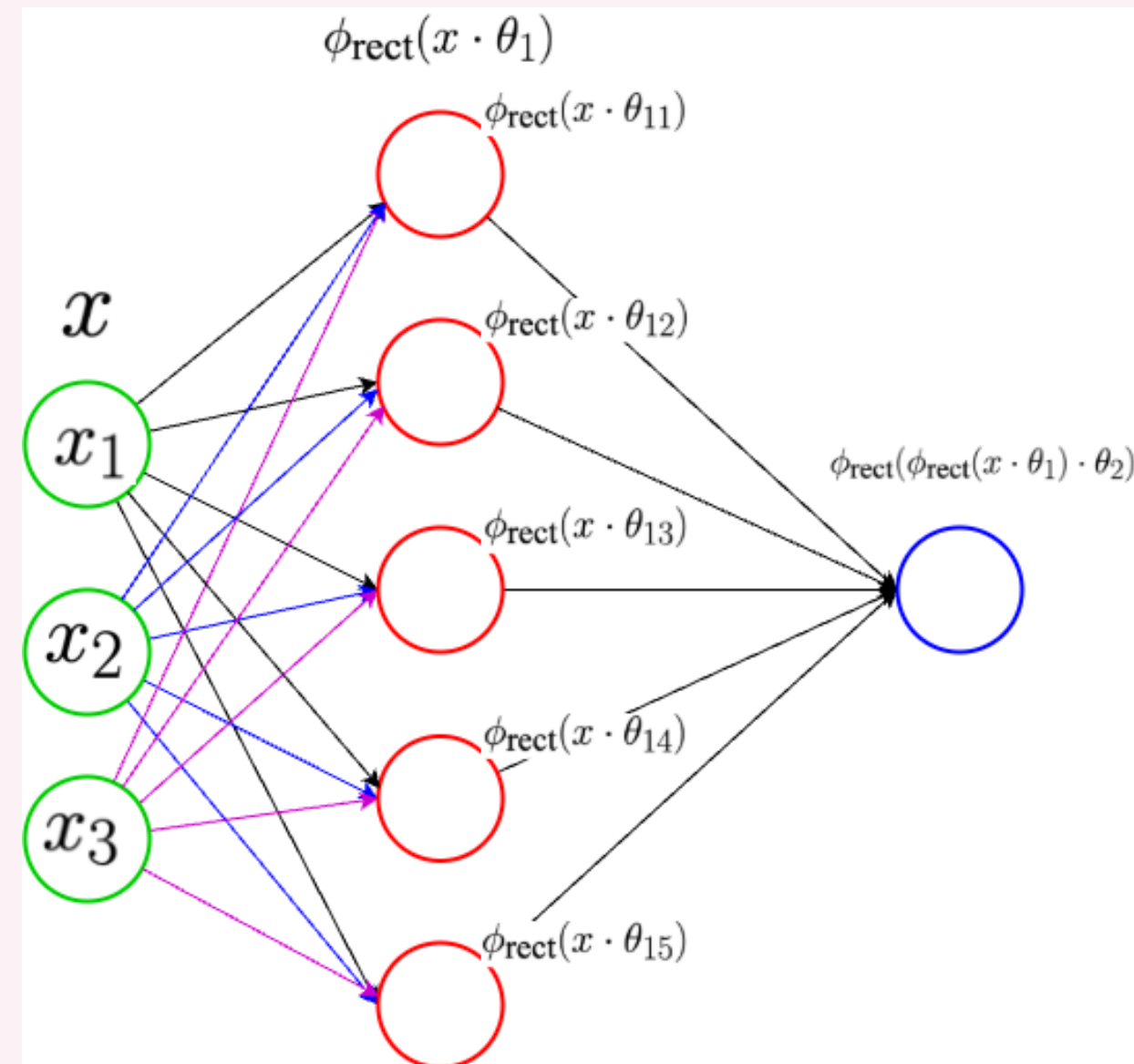
Properties of Deep Rectified Networks

Definition 3. Given K weight matrices $(\theta_k)_{k \leq K}$ with $n_k = \dim(\text{vec}(\theta_k))$ and $n = \sum_{k=1}^K n_k$, the output y of a deep rectified feedforward network with a linear output layer is:

$$y = \phi_{\text{rect}}(\phi_{\text{rect}}(\cdots \phi_{\text{rect}}(x \cdot \theta_1) \cdots \theta_{K-1}) \cdot \theta_K),$$

where

- x is the input to the model, a high-dimensional vector
- ϕ_{rect} is the rectified elementwise activation function, the positive part $(z_i)_i \mapsto (\max(z_i, 0))_i$.
- vec reshapes a matrix into a vector.



Rectified Feedforward Network with Three Linear Output Layers

Non-Negative Homogeneity and Scale Transformations

Definition 4. A given a function ϕ is non-negative homogeneous if

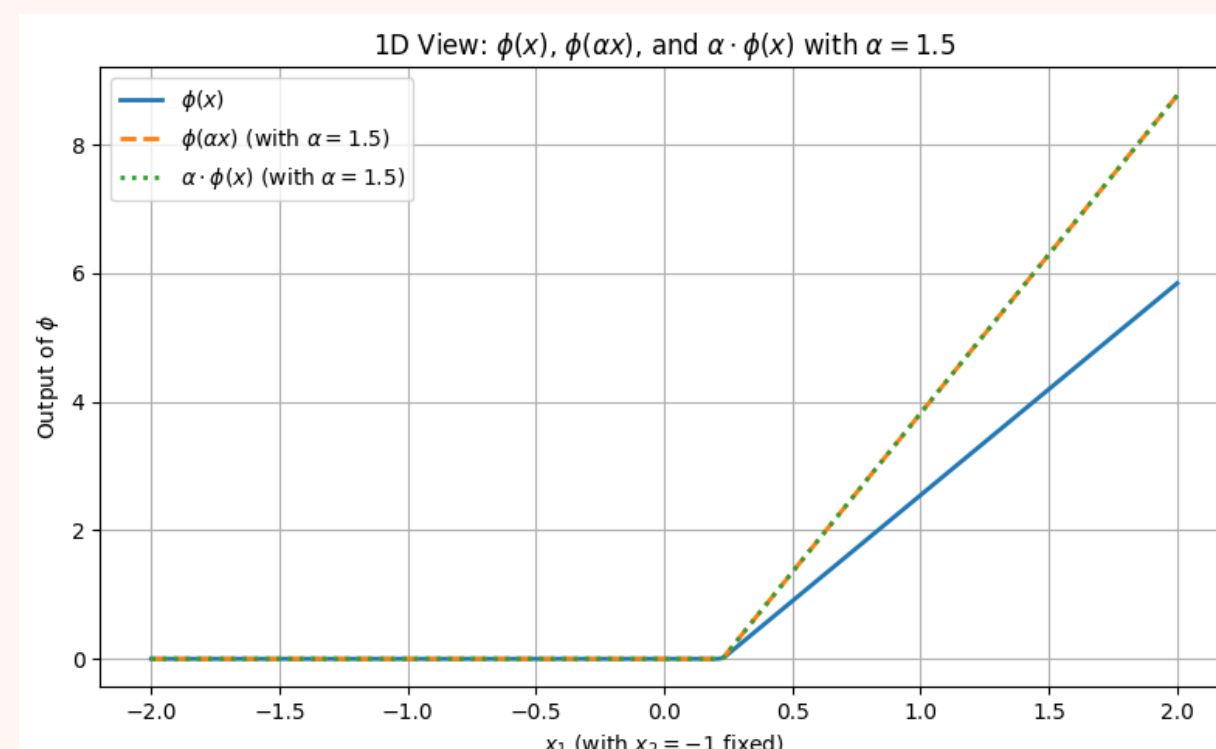
$$\forall (z, \alpha) \in \mathbb{R} \times \mathbb{R}^+, \quad \phi(\alpha z) = \alpha \phi(z)$$

Theorem 1. The rectified function $\phi_{\text{rect}}(x) = \max(x, 0)$ is non-negative homogeneous.

Definition 5. For a single hidden layer rectifier feedforward network we define the family of transformations

$$T_\alpha : (\theta_1, \theta_2) \mapsto (\alpha \theta_1, \alpha^{-1} \theta_2)$$

which we refer to as a α -scale transformation.



Rectified Functions Comparison

ϵ -flatness has volume under reparametrization

Theorem 2: For a one-hidden layer rectified neural network of the form $y = \phi_{\text{rect}}(x \cdot \theta_1) \cdot \theta_2$, and a minimum $\theta = (\theta_1, \theta_2)$, with $\theta_1, \theta_2 \neq 0$, $\forall \epsilon > 0$, $C(L, \theta, \epsilon)$ has an infinity volume.

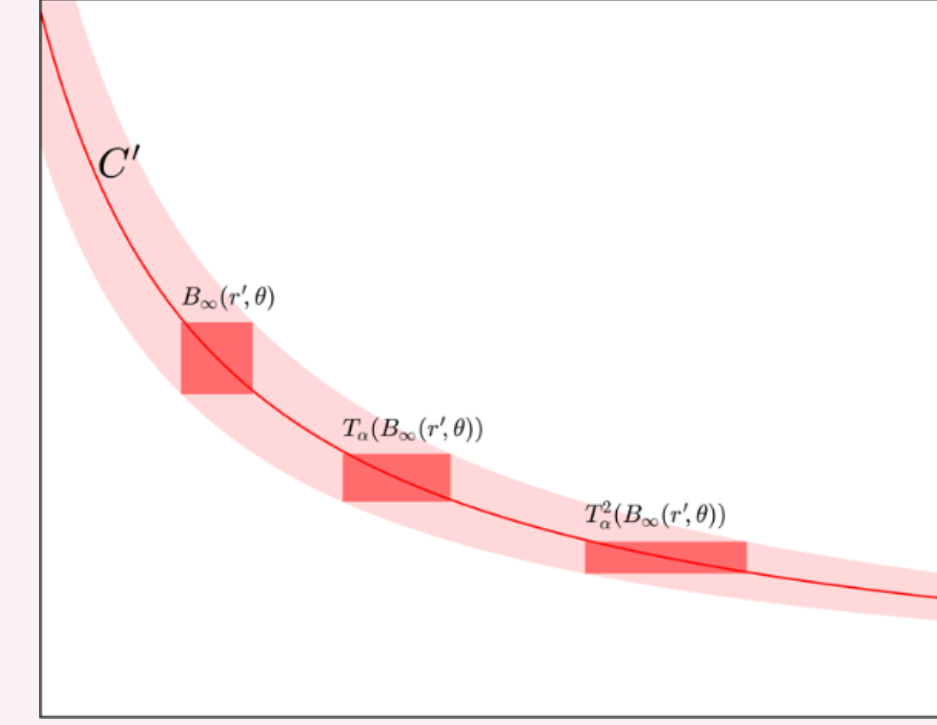
Proof:

Consider r such that the $\mathcal{L}_\infty$ ball $B_\infty(r, \theta) \subseteq C(L, \theta, \epsilon)$, with volume $v = (2r)^{n_1+n_2}$. The transformation $T_\alpha : (\theta_1, \theta_2) \mapsto (\alpha \theta_1, \alpha^{-1} \theta_2)$ has Jacobian Determinant $\alpha^{n_1-n_2}$.

- If $n_1 \neq n_2$: The volume of $(T_\alpha(C(L, \theta, \epsilon))) >$ the volume of $(T_\alpha(B_\infty(r, \theta))) = v \cdot \alpha^{n_1-n_2} \xrightarrow{\text{adjust } \alpha} \infty$.

- If $n_1 = n_2$: Let $C' = \bigcup_{\alpha' > 0} T_{\alpha'}(B_\infty(r, \theta))$, it's connected. Let $\alpha' = 2 \frac{\|\theta_1\|_\infty + r}{\|\theta_1\|_\infty - r}$, then $T_{\alpha'}(B_\infty(r, \theta)) \cap (B_\infty(r, \theta)) = \emptyset$.

Thus, $T_\alpha^k = T_{\alpha^k} \implies \{T_{\alpha^k}(B_\infty(r, \theta))\}_{k=1}^\infty$ is pairwise disjoint
 $\implies$ The volume of C' is lower bounded by $\sum_1^\infty v = \infty$.



Hessian-Based Measures

Theorem 3 (Gradient and Hessian transformation under reparameterization) Let $T_\alpha(\theta_1, \theta_2) = (\alpha \theta_1, \alpha^{-1} \theta_2)$. Then:

$$\nabla L(T_\alpha(\theta)) = \nabla L(\theta) D_\alpha, \quad \nabla^2 L(T_\alpha(\theta)) = D_\alpha^\top \nabla^2 L(\theta) D_\alpha$$

where $D_\alpha = \text{diag}(\alpha^{-1} I_{n_1}, \alpha I_{n_2})$.

Theorem 4 (Spectral norm inflation in 1-hidden-layer ReLU networks) Let $y = \phi_{\text{ReLU}}(x \cdot \theta_1) \cdot \theta_2$ be a one-hidden-layer ReLU network, and let $\theta = (\theta_1, \theta_2)$ be a critical point of L such that $\nabla^2 L(\theta) \neq 0$. Then:

$$\forall M > 0, \exists \alpha > 0 \text{ such that } \|\nabla^2 L(T_\alpha(\theta))\|_2 \geq M$$

Theorem 5 (Multi-directional sharpness in deep ReLU networks) Let

$$y = \phi_{\text{ReLU}}(\cdots \phi_{\text{ReLU}}(x \cdot \theta_1) \cdots) \cdot \theta_K$$

be a ReLU network with $K - 1$ hidden layers, and let $\theta = (\theta_k)_{k \leq K}$ be a critical point of L , such that $\text{rank}(\nabla^2 L(\theta)) = r$. Then:

$$\forall M > 0, \exists \{\alpha_k > 0\}_{k \leq K} \text{ with } \prod_{k=1}^K \alpha_k = 1 \text{ such that}$$

$$r - \min_{k \leq K} (n_k) \text{ eigenvalues of } \nabla^2 L(T_\alpha(\theta)) \geq M$$

These results show that curvature-based measures like spectral norm or Hessian eigenvalues are not invariant under reparameterization.

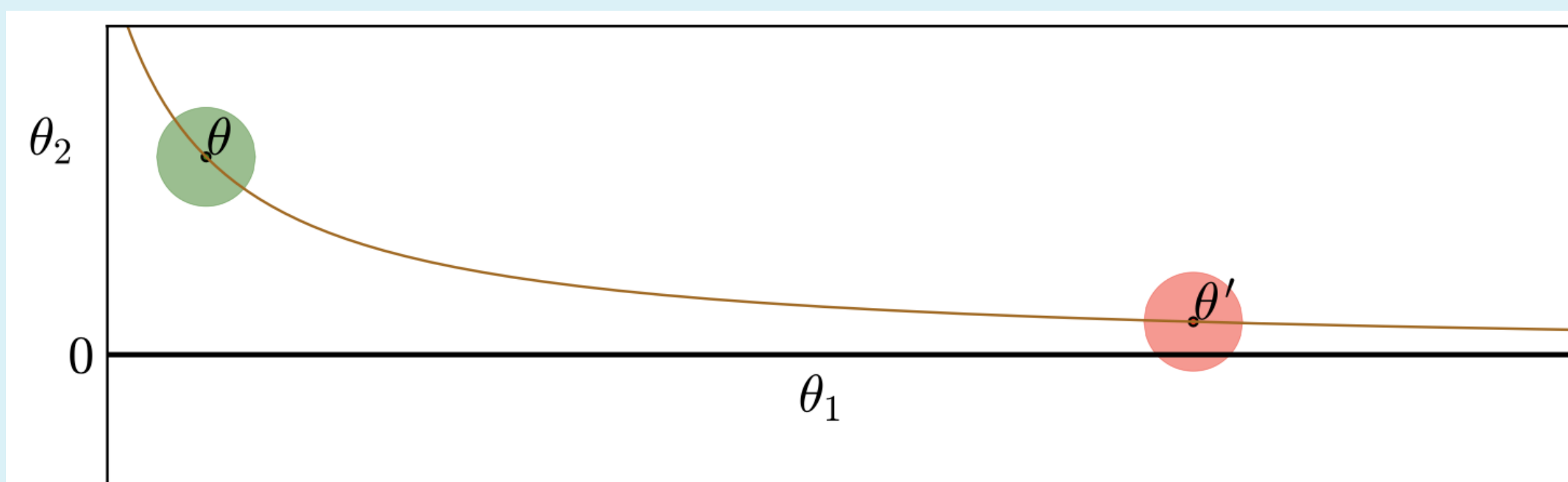
ϵ -sharpness

Definition 6 (ϵ -sharpness) For a minimum θ , and radius $\varepsilon > 0$:

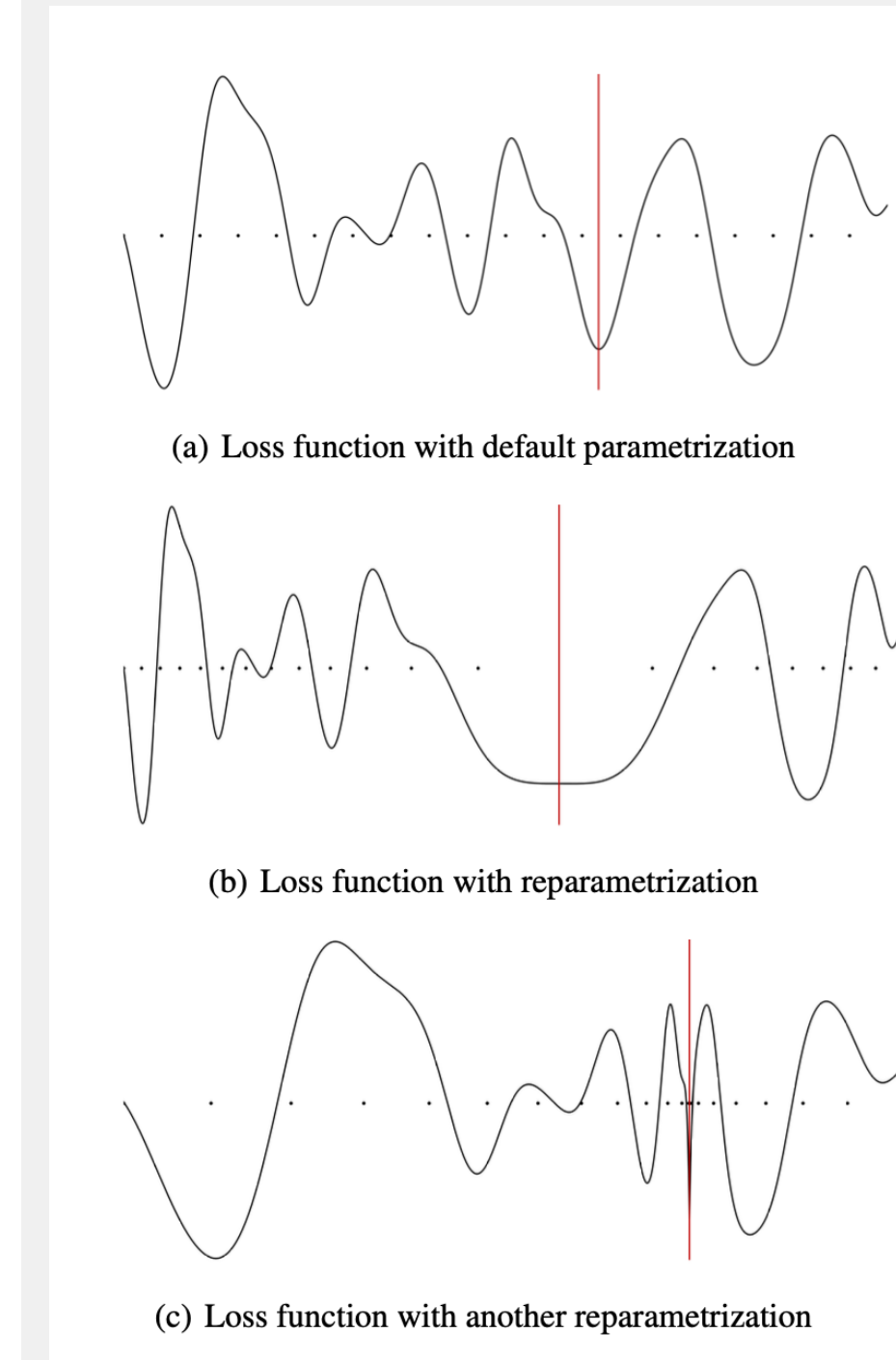
$$\varepsilon\text{-sharpness} = \frac{\max_{\theta' \in B_2(\varepsilon, \theta)} L(\theta') - L(\theta)}{1 + L(\theta)}$$

This quantity is not invariant under reparameterizations and can be made arbitrarily large, even for functions with identical generalization behavior.

Flatness metrics such as ε -sharpness do not capture function-intrinsic properties and are therefore unreliable as generalization measures.



Reparametrization and Input representation



Reparametrizations with $\eta = (|\theta - \hat{\theta}|^2 - b)^a (\theta - \hat{\theta})$ where $b \geq 0$, $a > -\frac{1}{2}$, and $\hat{\theta}$ is the red line

- Reparametrization:** Given loss function $L(\theta)$ and bijective function $g : \eta \mapsto \theta$ (e.g. $g(\eta) = \theta^3$), we write $L_\eta(\eta) = L(g(\eta))$

- g is bijective $\implies L$ is invariant under this reparametrization.
- After this reparametrization, ∇L and $\nabla^2 L$ changed according to the chain rule.

This means that we can arbitrarily stretch the loss function with bijective reparametrization, without changing the generalization behavior.

- Input Representation:** Given prediction function $f(x)$ and bijective function $\xi : u \mapsto x$, we write $f_\xi(u) = f(\xi(u))$

- ξ is bijective $\implies f$ is invariant under this input preprocessing.
- This input preprocessing changes the shape of L and the flatness/sharpness at L 's minimum.
- After this input preprocessing, the gradient of f w.r.t the input changes according to the chain rule.

This means that the input preprocessing also changes the loss function while keeping f invariant, without changing the generalization behavior..

Conclusion

The authors examine various definitions of flatness and argue that flat minima do not inherently generalize better than sharp ones. Flatness measures overlook key aspects of deep architectures, such as:

- Model symmetries and the non-identifiability of parameters, where different parameterizations can represent the same function.
- The fact that the loss surface geometry can be altered arbitrarily via reparameterization.

Thus, flatness must be redefined in context—tied to a model's specific parametrization and input representation.

Critical Reflections

- Theoretical results rely heavily on ReLU's positive homogeneity:

$$\text{ReLU}(\alpha x) = \alpha \cdot \text{ReLU}(x)$$

- making α -transformations particularly effective. Applicability to non-ReLU activations may vary.
- The use of the ℓ_∞ ball simplifies volume analysis, but the key idea—volume inflation via reparametrization—is not limited to this norm.
- The α -transformation:

$$T_\alpha(\theta_1, \theta_2) = (\alpha \theta_1, \alpha^{-1} \theta_2)$$

demonstrates that loss geometry is mutable. However, such transformations may not always be feasible in constrained, real-world training setups.

- While flatness may correlate with generalization in some settings, this paper shows such correlation is neither causal nor invariant under reparametrization.
- Geometry-based generalization heuristics require caution and context—they are neither sufficient nor universally applicable.

Acknowledgments and References

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Link to references